

Modelling Combinatorial Optimization Problems

Instances, feasible solutions and cost functions

Lecture 2

September 7, 2026

Combinatorial Optimization Problems

- Formally, a Combinatorial Optimization Problem (COP) is defined as a quadruple:

$$\mathcal{P} = (S, \Omega, f, \text{opt})$$

- Components:**

- S : Solution Space — finite or countable set of all candidate configurations
- $\Omega \subseteq S$: Feasible Region — set of solutions satisfying all problem constraints
- $f : \Omega \rightarrow \mathbb{R}$: Objective Function — maps each feasible solution to a real cost
- $\text{opt} \in \{\min, \max\}$: Optimization Goal

- Mathematical Goal:**

- Find an optimal feasible solution $x^* \in \Omega$ such that:

$$f(x^*) \leq f(x) \quad \forall x \in \Omega \quad (\text{for minimization})$$

Example 1— Traveling Salesperson Problem (TSP)

- **Problem Instance:** Given a complete weighted graph $G = (V, E, w)$
 - $V = \{1, 2, \dots, n\}$ represents n cities
 - $w(i, j) \in \mathbb{R}_{\geq 0}$ represents travel cost from city i to city j

- **Solution Space (S):** All permutations of the vertex set V :

$$S = S_n = \{\pi = (\pi_1, \pi_2, \dots, \pi_n) \mid \pi \text{ is a permutation of } V\} \quad (|S| = n!)$$

- **Feasible Region (Ω):** All valid Hamiltonian cycles (tours):

$$\Omega = S_n \quad (\text{every permutation forms a valid tour in a complete graph})$$

- **Objective Function (f):** Total tour length $f : \Omega \rightarrow \mathbb{R}^+$:

$$f(\pi) = \sum_{k=1}^{n-1} w(\pi_k, \pi_{k+1}) + w(\pi_n, \pi_1)$$

- **Optimization Goal (opt):** $\min_{\pi \in \Omega} f(\pi)$

Example 1— Traveling Salesperson Problem (TSP)

- **Formal Statement:**

Find $\pi^* \in S_n$ s.t. $f(\pi^*) = \min_{\pi \in S_n} f(\pi)$

	1	2	3	4
1	—	10	15	20
2	10	—	35	25
3	15	35	—	30
4	20	25	30	—

- **Toy Example:**

- $V = \{1, 2, 3, 4\}$ with symmetric matrix w
- Feasible set $\Omega = S_4$ contains $4! = 24$ permutations
- A possible tour: $\pi = (1, 3, 2, 4)$
- Tour path: $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$

$$f(\pi^*) = w(1, 3) + w(3, 2) + w(2, 4) + w(4, 1) = 15 + 35 + 25 + 20 = 95$$

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- **Toy Example:**

- $V = \{1, 2, 3, 4\}$ with symmetric matrix w
- Feasible set $\Omega = S_4$ contains $4! = 24$ permutations
- optimal tour: $\pi^* =$
- Tour path:

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2	10	—	35	25
3	15	35	—	30
4	20	25	30	—

$f(\pi^*) =$

Example 2— 0/1 Knapsack Problem

- **Problem Instance:** Set of n items with weights $w_i \in \mathbb{Z}^+$, values $v_i \in \mathbb{Z}^+$, and capacity $W \in \mathbb{Z}^+$
- **Solution Space (S):** All n -dimensional binary vectors:

$$S = \{y = (y_1, y_2, \dots, y_n) \mid y_i \in \{0, 1\}\} \quad (|S| = 2^n)$$

- **Feasible Region (Ω):** Bitstrings satisfying capacity constraint:

$$\Omega = \left\{ y \in S \mid \sum_{i=1}^n w_i y_i \leq W \right\}$$

- **Objective Function (f):** Total value accrued $f : \Omega \rightarrow \mathbb{R}^+$:

$$f(y) = \sum_{i=1}^n v_i y_i$$

- **Optimization Goal (opt):** $\max_{y \in \Omega} f(y)$

Example 2— 0/1 Knapsack Problem

- **Toy Example:** Capacity $W = 5$, items defined as (w_i, v_i) :

- Item 1: $w_1 = 2, v_1 = 3$ | Item 2: $w_2 = 3, v_2 = 4$ | Item 3:
 $w_3 = 4, v_3 = 5$

- **Evaluating Feasible Configurations (Ω):**

Vector y	Items Chosen	Total Weight	Feasible? ($y \in \Omega$)	Value $f(y)$
$(1, 0, 0)$	$\{1\}$	2	Yes	3
$(0, 1, 0)$	$\{2\}$	3	Yes	4
$(0, 0, 1)$	$\{3\}$	4	Yes	5
$(1, 1, 0)$	$\{1, 2\}$	$2 + 3 = 5$	Yes	7 (Optimal y^*)
$(1, 0, 1)$	$\{1, 3\}$	$2 + 4 = 6$	No (> 5)	—
$(0, 1, 1)$	$\{2, 3\}$	$3 + 4 = 7$	No (> 5)	—
$(1, 1, 1)$	$\{1, 2, 3\}$	$2 + 3 + 4 = 9$	No (> 5)	—

Role of Constraints in Combinatorial Optimization

- **Constraints** restrict candidate solutions S to valid configurations Ω .
- **Hard Constraints vs. Unconstrained Problems:**
 - *Constrained* ($\Omega \subset S$): Solutions must satisfy algebraic or structural rules (e.g., Knapsack capacity $\sum w_i y_i \leq W$).
 - *Unconstrained* ($\Omega = S$): Every candidate configuration in S is valid (e.g., TSP over a complete graph, Max-Cut).

- **Mathematical Representation with Penalty / Indicator Constraints:**

$$\max_{y \in S} f(y) \quad \text{subject to} \quad g_k(y) \leq 0, \quad k \in \{1, \dots, m\}$$

- Formally, $\Omega = \{y \in S \mid g_k(y) \leq 0, \forall k\}$. If $y \notin \Omega$, y is **infeasible**.

Example 1 — TSP: Binary ILP Formulation

- **Alternative COP Model:** Binary Integer Linear Program (0-1 ILP)
- **Decision Variables:** $x_{ij} \in \{0, 1\}$ ($x_{ij} = 1$ if edge (i, j) is in tour)

$$\min \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n c_{ij} x_{ij}$$

Subject to:

Example 1 — TSP: Binary ILP Formulation

- *Out-Degree Constraints:*

$$\sum_{\substack{j=1 \\ j \neq i}}^n x_{ij} = 1 \quad \forall i \in V$$

- *In-Degree Constraints:*

$$\sum_{\substack{i=1 \\ i \neq j}}^n x_{ij} = 1 \quad \forall j \in V$$

- *Integrality Constraints:*

$$x_{ij} \in \{0, 1\} \quad \forall i, j \in V, i \neq j$$

- *Subtour Elimination Constraints (Dantzig-Fulkerson-Johnson):*

$$\sum_{i \in Q} \sum_{\substack{j \in Q \\ j \neq i}} x_{ij} \leq |Q| - 1$$

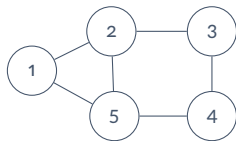
$$\forall Q \subset V, 2 \leq |Q| \leq n - 1$$

Example 3— Maximum Cut Problem (Max-Cut)

- **Problem Instance:** Undirected graph $G = (V, E)$ with edge weights $w_{ij} \geq 0$.
- **Solution Space (S):** All binary partition vectors for $V = \{1, \dots, n\}$:

$$S = \{y \in \{0, 1\}^n\} \quad (|S| = 2^n)$$

- **Feasible Region (Ω):** Unconstrained ($\Omega = S$).
Vertices are partitioned into $S_0 = \{i \mid y_i = 0\}$
and $S_1 = \{i \mid y_i = 1\}$.

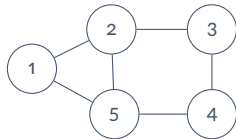


Example 3— Maximum Cut Problem (Max-Cut)

- **Objective Function (f):** Sum of cut edge weights:

$$f(y) = \sum_{(i,j) \in E} w_{ij}(y_i - y_j)^2 = \sum_{(i,j) \in E} w_{ij}(y_i + y_j - 2y_i y_j)$$

- **Goal:** $\max_{y \in \Omega} f(y)$

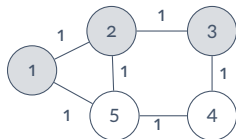


Example 3— Max-Cut Evaluation

- **Evaluating Partition $y = (1, 1, 1, 0, 0)$:**
 - $S_1 = \{1, 2, 3\}, S_0 = \{4, 5\}$
 - Edges crossing cut: $(3, 4), (1, 5), (2, 5)$
- **Cost Calculation ($w_{ij} = 1$):**

$$f(y) = \sum_{(i,j) \in E} w_{ij} |y_i - y_j| = 1 + 1 + 1 = 3$$

- **Optimal Solution y^* :** $y^* = (1, 0, 1, 0, 1)$ cuts 5 edges, yielding $f(y^*) = 5$.



Example 4— Job Scheduling ($P \parallel C_{\max}$)

- **Problem Instance:** n jobs with processing times $p_j \in \mathbb{R}^+$ and m identical parallel machines.
- **Solution Space (S):** All assignments of n jobs to m machines:

$$S = \{\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n) \mid \sigma_j \in \{1, \dots, m\}\} \quad (|S| = m^n)$$

- **Feasible Region (Ω):** Unconstrained ($\Omega = S$).
- **Objective Function (f):** Minimize the ****Makespan**** $C_{\max}$ (completion time of the last busy machine):

$$f(\sigma) = \max_{k \in \{1, \dots, m\}} \left(\sum_{j: \sigma_j = k} p_j \right)$$

- **Optimization Goal (opt):** $\min_{\sigma \in \Omega} f(\sigma)$

Example 4— Job Scheduling Numerical Example

- **Instance:** $n = 4$ jobs with processing times $p = (3, 5, 2, 6)$ on $m = 2$ identical machines.
- **Candidate Assignments ($\sigma \in \Omega$):**

Assignment σ	Machine 1 Jobs	Machine 2 Jobs	Loads (L_1, L_2)	Makespan $f(\sigma) = \max(L_1, L_2)$
(1, 1, 1, 2)	{1, 2, 3}	{4}	$(3 + 5 + 2, 6) = (10, 6)$	10
(1, 2, 2, 1)	{1, 4}	{2, 3}	$(3 + 6, 5 + 2) = (9, 7)$	9
(1, 2, 1, 2)	{1, 3}	{2, 4}	$(3 + 2, 5 + 6) = (5, 11)$	11
(2, 1, 1, 2)	{2, 3}	{1, 4}	$(5 + 2, 3 + 6) = (7, 9)$	9

- **Optimal Solution (σ^*):** $\sigma^* = (1, 2, 2, 1)$ with optimal makespan $f(\sigma^*) = 9$.

Example 5— Resource Allocation Problem

- **Problem Instance:** n activities, m resources with total availabilities B_k . Activity j consumes a_{kj} units of resource k and yields utility $u_j(x_j)$.
- **Solution Space (S):** Discrete allocation quantities for each activity:

$$S = \{x = (x_1, x_2, \dots, x_n) \mid x_j \in \{0, 1, \dots, U_j\}\}$$

- **Feasible Region (Ω):** Allocations respecting all capacity constraints:

$$\Omega = \left\{ x \in S \mid \sum_{j=1}^n a_{kj} x_j \leq B_k \quad \forall k \in \{1, \dots, m\} \right\}$$

- **Objective Function (f):** Total utility across activities: $f(x) = \sum_{j=1}^n u_j(x_j)$
- **Optimization Goal (opt):** $\max_{x \in \Omega} f(x)$

Example 5— Resource Allocation Numerical Example

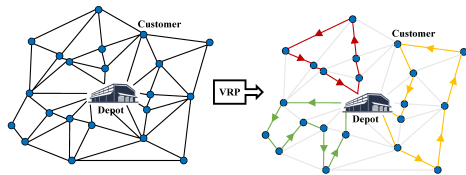
- **Instance:** $n = 3$ projects, $m = 1$ resource (budget $B = 10$). Decision $x_j \in \{0, 1, 2\}$.
- **Resource Matrix & Utility Table:**
 - Unit costs $a = (a_1, a_2, a_3) = (2, 3, 4)$
 - Utility functions $u_j(x_j) = v_j \cdot x_j$ with values $v = (4, 7, 9)$
- **Evaluating Configurations ($x \in S$):**

Allocation x	Total Cost $\sum a_j x_j$	Feasible? ($x \in \Omega$)	Total Utility $f(x) = \sum v_j x_j$
(1, 1, 1)	$2(1) + 3(1) + 4(1) = 9$	Yes (≤ 10)	$4(1) + 7(1) + 9(1) = 20$
(0, 2, 1)	$2(0) + 3(2) + 4(1) = 10$	Yes (≤ 10)	$4(0) + 7(2) + 9(1) = 23$ (x^*)
(1, 0, 2)	$2(1) + 3(0) + 4(2) = 10$	Yes (≤ 10)	$4(1) + 7(0) + 9(2) = 22$
(0, 1, 2)	$2(0) + 3(1) + 4(2) = 11$	No (> 10)	—

Example 6— Vehicle Routing Problem (VRP)

- **Informal Problem Definition:**

- A fleet of K identical vehicles located at a central **depot** must serve a set of geographically dispersed customers.
- **Goal:** Design optimal routes for all K vehicles to serve all customers at minimum total transit cost.



- **Key Operational Constraints:**

- Every vehicle starts and ends its tour at the central depot.
- Every customer is visited exactly once by a single vehicle.

Example 6— Basic ("Vanilla") VRP Model

- **Problem Instance:** Complete directed graph $G = (V, E, d)$ with K vehicles
 - $V = \{0, 1, \dots, n\}$ where 0 represents the central depot and $V \setminus \{0\}$ are customers
 - $d_{ij} \in \mathbb{R}_{\geq 0}$ represents distance/travel cost between locations i and j
- **Solution Space (S):** Binary assignment variables $x_{ij}^k \in \{0, 1\}$ for each vehicle k :

$$S = \left\{ x = (x_{ij}^k)_{i,j \in V, k \in \{1, \dots, K\}} \mid x_{ij}^k \in \{0, 1\}, i \neq j \right\}$$

- **Feasible Region (Ω):** Assignments satisfying structural flow constraints:
 - Customer Visitation: $\sum_{k=1}^K \sum_{j \in V, j \neq i} x_{ij}^k = 1 \quad \forall i \in V \setminus \{0\}$
 - Flow Conservation: $\sum_{j \neq i} x_{ij}^k = \sum_{j \neq i} x_{ji}^k \quad \forall i \in V, \forall k$
 - Depot Departure / Return: $\sum_{j \in V \setminus \{0\}} x_{0j}^k = 1, \quad \sum_{i \in V \setminus \{0\}} x_{io}^k = 1 \quad \forall k$
- **Objective Function (f):** $f(x) = \sum_{k=1}^K \sum_{i \in V} \sum_{j \in V, j \neq i} d_{ij} x_{ij}^k$ (**Goal:** $\min_{x \in \Omega} f(x)$)

Example 6— "Vanilla" VRP: Binary ILP

- **Binary Decision Variable:** $x_{ij}^k = 1$ if vehicle k travels directly from node i to j ; 0 otherwise.

$$\min \sum_{k=1}^K \sum_{i \in V} \sum_{\substack{j \in V \\ j \neq i}} d_{ij} x_{ij}^k$$

Example 6— "Vanilla" VRP: Binary ILP

Subject to:

- *Single Customer Visit:*

$$\sum_{k=1}^K \sum_{\substack{j \in V \\ j \neq i}} x_{ij}^k = 1 \quad \forall i \in V \setminus \{o\}$$

- *Flow Balance at Nodes:*

$$\sum_{\substack{j \in V \\ j \neq i}} x_{ij}^k - \sum_{\substack{j \in V \\ j \neq i}} x_{ji}^k = 0 \quad \forall i \in V, \forall k$$

- *Depot Departure / Arrival:*

$$\sum_{j \in V \setminus \{o\}} x_{oj}^k = 1 \quad \forall k \in \{1, \dots, K\}$$

$$\sum_{i \in V \setminus \{o\}} x_{io}^k = 1 \quad \forall k \in \{1, \dots, K\}$$

Example 7 — Capacitated VRP (CVRP)

- **Problem Instance Extension:**

- Vehicle capacity: $Q \in \mathbb{Z}^+$ (uniform across all K vehicles)
- Customer demand: $D_i \in \mathbb{Z}^+$ for each customer $i \in V \setminus \{o\}$ (with $D_o = 0$)

- **Auxiliary Decision Variables:**

- $u_i^k \geq 0$: Cumulative load carried by vehicle k upon leaving node i

- **Feasible Region (Ω_{CVRP}):** Extends Vanilla VRP constraints with capacity bounds:

$$\Omega_{\text{CVRP}} = \left\{ x \in \Omega_{\text{VRP}}, u \mid \sum_{i \in V \setminus \{o\}} D_i \sum_{j \neq i} x_{ij}^k \leq Q \quad \forall k \right\}$$

- **Subtour Elimination & Capacity Tracking (MTZ-style):**

$$u_i^k - u_j^k + Qx_{ij}^k \leq Q - D_j \quad \forall i, j \in V \setminus \{o\}, i \neq j, \forall k$$

$$D_i \leq u_i^k \leq Q \quad \forall i \in V \setminus \{o\}, \forall k$$

CVRP Formulation

$$\min \sum_{k=1}^K \sum_{i \in V} \sum_{\substack{j \in V \\ j \neq i}} d_{ij} x_{ij}^k$$

Subject to:

- Customer Coverage:

$$\sum_{k=1}^K \sum_{j \neq i} x_{ij}^k = 1 \quad \forall i \in V \setminus \{o\}$$

- Depot Start/End:

$$\sum_{j \neq o} x_{oj}^k = 1, \quad \sum_{i \neq o} x_{io}^k = 1 \quad \forall k$$

- Capacity & Subtour Elimination:

$$u_i^k - u_j^k + Qx_{ij}^k \leq Q - D_j \quad \forall i, j \neq o, i \neq j$$

- Load Bounds:

$$D_i \leq u_i^k \leq Q \quad \forall i \in V \setminus \{o\}, \forall k$$